Ceramics International 50 (2024) 21788–21799

Contents lists available at ScienceDirect

Ceramics International

journal homepage: www.elsevier.com/locate/ceramint

Microstructures and mechanical properties of copper-to-glass laser transmission welding employing a nanosecond pulsed laser

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ARTICLE INFO

Handling editor: Dr P. Vincenzini

Keywords:
Laser transmission welding
Glass/copper bonding
Microstructures
Cu^{x+}-O-Si eutectic compounds
Bond strength

ABSTRACT

This study focuses on the interfacial characteristics of glass/copper welds achieved through laser transmission welding, utilizing a nanosecond pulsed laser. The investigation explores key processing parameters, specifically laser line energy intensity and total scan numbers, and their impacts on weld performance. Through meticulous analysis covering bond strength, cross-sectional surface morphologies, micro- and nano-structures, phase formations, and elemental compositions, the study systematically addresses the weld formation mechanism. The research elucidates potential reasons for diverse bonding characteristics, emphasizing the formation of Cu^{x+}-O-Si compounds (primarily Cu^{+}-O-Si and Cu^{2+}-O-Si) at the glass/copper interfacial region, as revealed by the XPS spectrum and HR-TEM images. These findings highlight redox reactions facilitated by strong intra-mixing through diffusion and convection during welding. The shear force separation test unveils an impressive maximum bond strength of 34.9 MPa, surpassing previously reported results, despite the indication of a brittle fracture in the separated surface morphologies. The correlation between a larger weld zone, abundant eutectic Cu^{x+}-O-Si compounds, and a uniformly re-solidified glass region is closely linked to the resulting high weld strength. This study underscores the direct relationship between laser irradiating energy and the extent of the weld zone, while the number of repeated scans proved to be an effective approach for regulating the elemental composition and microstructures in the weld. Overall, these findings contribute valuable insights into the weld formation mechanism and offer practical considerations for optimizing glass/copper welds through laser transmission welding.

1. Introduction

The fusion of glass and metal is widely employed across various fields, including electronic packages [1], micro-optoelectromechanical systems (MOEMS) [2], and intricate miniature sensors [3]. Using a pulsed laser for bonding glass and metal presents an attractive alternative to conventional welding techniques like anodic bonding [4], diffusion bonding [5], and adhesive bonding [6]. This approach offers notable benefits such as reduced defect rates and improved welding strength. Notably, nanosecond (ns) lasers offer several advantages over femtosecond (fs) and picosecond (ps) lasers, including stable output, lightweight structure, cost-effectiveness, and minimal operational and maintenance requirements [7,8], making them the preferred choice in the laser processing industry.

Recently, ultrashort pulsed lasers, including fs and ps lasers, have garnered significant attention for welding applications due to their exceptional precision in delivering welding energy within a defined range. These lasers induce minimal heat-affected zones, making them suitable for precise and efficient welding processes. For example, Zhang et al. [9] successfully employed an fs laser with a repetition rate of 1 kHz and a pulse duration of 160 fs to weld copper and glass samples. Similarly, Li et al. [10] investigated Cu/SiO_2 joints using an ultrashort pulsed laser with a wavelength of 1030 nm and a pulse duration of 300 fs, demonstrating the potential of ultrashort pulsed lasers for joining glass and copper. However, despite their advantages, challenges persist in their utilization, including ensuring the gap between the glass and metal falls within a few micrometer range, operating in demanding environments, and addressing the complexity of the systems, which can lead to elevated costs. Consequently, it is imperative to meticulously evaluate the appropriate bonding method for metal-to-glass applications.

Utilizing longer pulse lasers offers a cost-effective alternative. For instance, Huo et al. [7] proposed the use of an ns laser to weld stainless steel 304 and silicate glass, resulting in a joint exhibiting superior bonding quality with a strength of 16.39 MPa. Similarly, Li et al. [8]

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https://doi.org/10.1016/j.ceramint.2024.03.291
Received 17 January 2024; Received in revised form 15 March 2024; Accepted 23 March 2024
Available online 2 April 2024
0272-8842/

investigated the bonding of stainless steel to soda-lime glass using an ns laser, achieving a maximum shear strength of 9.74 MPa. Furthermore, Li et al. [11] conducted laser transmission welding (LTW) between titanium alloy and borosilicate glass using a long-pulsed Nd:YAG laser, with the highest bonding strength reaching 6.49 MPa. Although these studies demonstrated the feasibility of welding glass and metal using nanosecond pulsed lasers, achieving optimal bonding strength was limited by challenges in determining an appropriate range of welding parameters to ensure proper weld quality.

On the other hand, evaluating welding performance typically involves examining the formation of new compounds resulting from chemical bonds or reactions between elements during the welding process. In the specific context of millisecond laser welding of glass to aluminum, Min et al. [12] utilized X-ray diffractometer (XRD) analysis with EDS to illustrate and identify the formation of the new compound $Al_2Si_4O_{10}$. Similarly, in the case of laser transmission welding of glass to copper, Feng et al. [13] identified the presence of the Cu–O–Si compound and Cu_2O at the copper/glass interface using scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and X-ray photoelectron spectroscopy (XPS) analyses, underscoring their critical role in enhancing bonding strength. A recent study by Wei et al. [14] also emphasized that the formation of a new amorphous zone combining Cu, Si, and O elements significantly contributed to a substantial enhancement in welding quality. It is noteworthy that while micro-scale examination tools such as SEM, transmission electron microscope (TEM), XRD, and XPS have been employed to characterize the weld, the majority of existing studies have focused on cleaved surfaces, either from the glass side or the copper side.

Regarding the influence of commonly used laser welding processing parameters on the bonding quality of metal/glass joints, Wang et al. [15] investigated the bonding of glass and 304 stainless steel using an fs laser. They examined the effects of laser pulse energy, welding speed, focal position, and laser frequency on weld quality, achieving a high shear strength of 8.79 MPa. Utsumi et al. [16] utilized an ns laser system to weld borosilicate glass and copper, achieving a maximum shear strength of 690 kPa with a pulse energy of 300 μJ. Similarly, Huo et al. [7] conducted welding between silica glass and stainless steel using an ns laser, investigating the impact of laser power, frequency, and scanning speed on bonding quality, and obtained an elevated bonding strength of 16.39 MPa.

In addition to examining the influence of commonly discussed processing parameters such as laser power, pulse repetition rate, and laser beam scanning speed on weld characteristics, microstructures of the weld can be finely tuned through suitable post-processing methods for microstructure modification, thereby improving weld quality. Beyond conventional processing, the laser itself serves as an excellent tool for localized post-processing, leveraging its exceptional patterning capabilities. Within the laser welding infrastructure, post-processing can be seamlessly integrated without altering processing configurations, achieved through repetitive laser scans. For instance, Feng et al. [13] demonstrated that an appropriate number of post-laser scans could enhance bonding strength by reducing micro-defects within the fusion zone, resulting in a maximum bonding shear strength of 33.4 MPa.

Moreover, recent studies have predominantly concentrated on assessing the influence of processing parameters on bonding quality by analyzing results over a large welding area formed through multiple overlapped line scans. This approach presents challenges in comprehending the weld of an individual scanned line, despite the importance of line welding in applications like sealing in optoelectronic and MOEMS devices, as well as metallization on printed circuit boards.

In this study, we assess the microstructure and mechanical properties of copper/glass joints utilizing an ns pulsed laser. Our objective is to determine the optimal range of processing parameters tailored to the characteristics of an ns laser. To gain insights into the impact of welding parameters on weld morphologies within and around the periphery of the weld seam, we employ SEM, high-resolution X-ray photoelectron spectroscopy (HR-XPS), and high-resolution transmission electron microscopy (HR-TEM) to characterize the microstructures, nanostructures, elemental compositions, and phase configurations of the weld seam before separation, thus eliminating the effects of separation force. Furthermore, the mechanical properties of the welded joints are characterized by shear strength, correlating with laser line energy intensity, total number of scans, geometries of the weld seam interface, and fracture surface morphology. This comprehensive analysis aims to advance the development and application of nanosecond laser technology for bonding transparent materials and metals.

2. Materials and experimental procedure

Experimental specimens comprised an EAGLE XG® glass substrate, measuring 25 mm × 25 mm × 0.7 mm, and a C1100 copper plate, with dimensions of 50 mm × 25 mm × 3 mm. The chemical compositions of these materials are detailed in Table 1. Furthermore, assessing bonding quality involves understanding the temperature changes during the bonding procedure, which play a critical role. Various physical properties, such as the coefficient of thermal expansion, thermal conductivity, melting point, and density of the two base materials, are presented in Table 2.

The initial surface roughness of the as-purchased copper plates varies from one another, approximately with a Ra of 1.5 μm, which was relatively coarse for direct use in bonding. Consequently, the initial C1100 copper plates underwent a polishing process before being utilized in the LTW process. Various grit sizes of sandpaper, specifically 600, 800, 1200, 2000, and 4000, were sequentially employed to grind the copper plates until an average surface roughness of approximately a Ra of 0.25 μm was achieved. The as-purchased EAGLE XG® glass substrate exhibited a surface roughness of about Ra = 0.15 μm. Both the glass and copper surfaces underwent a thorough cleansing regimen involving ethyl alcohol (C_2H_5OH) and were then subjected to 25 min of ultrasonic agitation, followed by a meticulous acetone rinse. Subsequently, the surfaces underwent a deionized water rinse and were carefully dried using nitrogen gas. The assembly of the two welding surfaces was facilitated through clamping jigs with a clamping force of 70 N. This arrangement ensured not only superior bonding outcomes but also maintained a minimal separation distance between the two surfaces designated for welding.

In Fig. 1(a), the experimental arrangement for the ns-laser system employed in this study is presented. A pulsed fiber laser system (YLP-1-100-20-20, IPG) with a maximum average laser power of 20 W, a pulse width of 100 ns, and a wavelength of 1070 nm was utilized for the laser bonding experiments. The welding system comprised an X-Y worktable, a focusing lens, and laser beam delivery mirrors. The focusing lens, with a focal distance of 290 mm, directed the laser beam to a spot with a size of 40 μm on its focal plane. Significantly, the manual Z-Axis 800 mm pillar allowed for vertical adjustment of the plane of focus. The Galvanometer scanner was securely affixed to a plate transversely attached to the pillar, enabling precise control of the laser spot's position along the Z-axis. Furthermore, a coaxial CCD camera monitored the position of the laser light beam on the focal plane.

For examination of the surface morphology of the weld seam on the cross-sectional plane, the welded samples underwent cutting using an IsoMet™ 4000 Linear Precision Saw and EXTEC Diamond Wafering Blades. Subsequently, the cut samples were polished with 0.5 μm diamond oil utilizing a Buehler Ecomet 30 Semi-Automatic Grinder Polisher to assess the surface topography of the cross-sectional region. To eliminate any dirt or impurities, the polished samples underwent a thorough cleaning process involving ethyl alcohol (C_2H_5OH) and a 15-min ultrasonic wash. After the cleaning procedure, cross-sectional surface images were obtained using a CLSM (Keyence VK-X1000) and an SEM (JOEL, JSM-7000F).

Apart from surface images, the XPS was employed to analyze the elemental compositions in the Cu/SiO_2 weld zone. The technique of the

Table 1
Chemical composition of EAGLE XG® glass and C1100 copper.

Material	Element (wt.%)						
	SiO_2	Al_2O_3	B_2O_3	MgO	CaO	SrO	BaO
Free-alkali glass	59.7	17.11	12.27	1.39	7.33	0.81	0.01
	Cu	O	Pb	S	Bi	Sb	As
C1100 copper	≥99.9	≤0.06	≤0.005	≤0.005	≤0.002	≤0.002	≤0.002

Table 2
Physical properties of C1100 copper and EAGLE XG® glass.

Material properties	C1100 copper	EAGLE XG® glass
Coefficient of thermal expansion (1/K)	16.9×10^{-6}	3.4×10^{-6}
Thermal conductivity ((W/(m.K))	391.1 (20 °C)	1.1 (25 °C)
Melting point (°C)	1083	669 (Strain Point) 722 (Annealing point) 971 (Softening Point) 1293 (Working point)
Density (g/cm^3)	8.96 (20 °C)	2.38

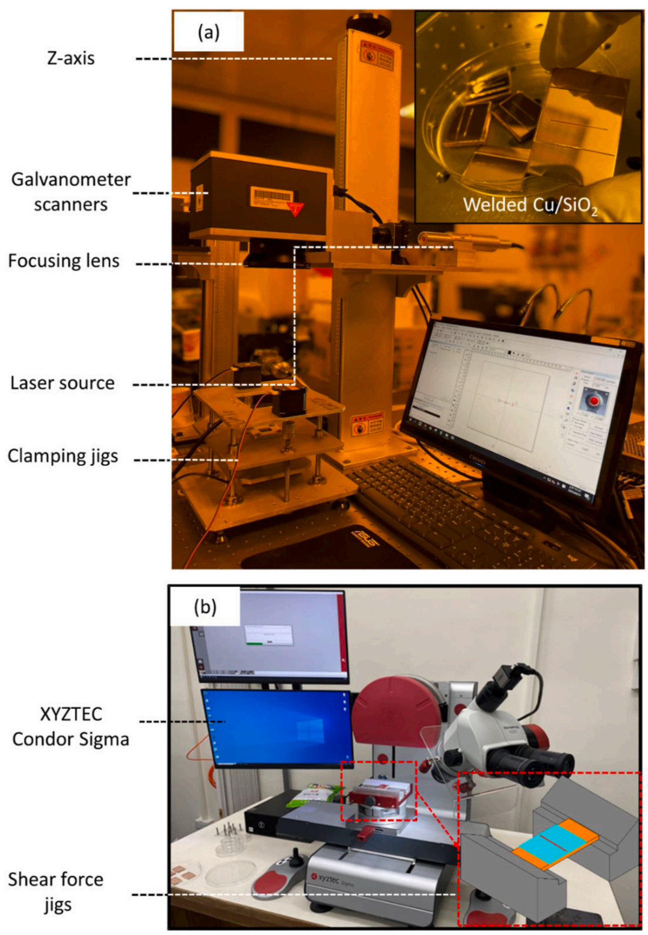

Fig. 1. Displays photographs illustrating the experimental setup for laser transmission welding (a) and the shear force separation measurement equipment (b) utilized in this study.

dual-beam focused ion beam (FIB, Versa 3D) was utilized to section samples around the welded glass/copper interface, enabling the examination of nanoscaled morphologies and structures at the interfacial level. HR-TEM (JEOL, JEM-2100) was employed to identify any potentially newly formed phases within the weld zone.

In this investigation, the bonding strength of the weld was assessed by determining the fracture point of the weld's maximum sustainable separation shear force, measured utilizing an XYZTEC Condor Sigma Series: Condor Sigma. This device, configurable with six sensors and a maximum shear force of 1000 N, holds the promise of enhancing existing standards. Fig. 1(b) illustrates the measurement tool employed for determining the maximum shear force.

3. Results and discussion

3.1. Key considerations in LTW of copper and glass

The morphology of welded joints between glass and copper, as well as the ultimate welding quality, are not solely determined by the initial roughness of the glass and copper surfaces. Rather, they are also significantly influenced by various laser processing parameters employed during the LTW. These crucial parameters include power, P, pulse repetition frequency, f, spot size, scanning speed, v, focal plane position, number of scans, N, and others.

However, it is important to note that, in the actual welding process where light interacts with the material, these parameters exhibit a combined dynamic effect. This results in their interdependence rather than mathematical independence. When considering the material's non-linear effects throughout processing, including nonlinear light-material interactions, temperature-dependent property variations, phase-transition-induced structural and property differences, and other intricate factors, the involved phenomena take on a notably high level of non-linearity. This complexity is further compounded, especially within the short duration of ns laser pulses. The intricacies arising from the interplay of these factors contribute to the challenging nature of understanding and controlling the welding process, demanding a comprehensive approach for achieving optimal results.

To streamline the intricacies and delineate a suitable range of welding parameters (processing window) for our current system in this study, we maintained certain constants. Specifically, we set $f = 20$ kHz, kept the focal plane position fixed at $P_f = -0.5$ mm (the negative sign denoting below the welding interface and beneath the copper surface) [17], and ensured a consistent jig clamping force (C_F) of 70 N. The variables subjected to variation were P, v, and N.

In our prior investigation [17], numerous test results highlighted the relatively narrow parameter window suitable for welding within the capabilities of the current laser system. Power levels below 15 W proved insufficient for effective welding while surpassing 16 W resulted in poor homogeneity and structural integrity issues. Consequently, this study opted for two power levels, namely $P = 15$ W and 16 W. Simultaneously, we delved into the interplay of scanning speed at two levels, $v = 35$ and $v = 40$ mm/s, respectively, combined with $f = 20$ kHz. This combination yielded a pulse overlapping factor of the beam spot higher than 95%. An often-utilized parameter that encapsulates incident laser power and scanning speed in welding is the line energy intensity, defined as $L = P/v$ (J/mm), signifying the irradiated laser energy per unit length along the welding direction. Additionally, we investigated the influence of the number of repeated scan times, functioning akin to post-processing, on the welded zone and its periphery. A judicious number of repeated scans may offer additional opportunities to facilitate inter-diffusion and chemical reactions, regulating microstructures around the copper/glass interface and inside the weld. The studied total scan numbers encompassed $N = 1$, 3, 4, 5, and 6.

Unlike ps laser and fs laser for glass/metal welding, glass can absorb laser energy direct through a non-linear absorption mechanism. Nd:YAG laser light is highly transparent to glass, with over 90% of it being transmitted. Consequently, in the ns laser welding of glass and copper, the primary joining mechanism involves the laser penetrating through the glass, being absorbed by the copper surface. This absorption leads to a temperature increase in copper, inducing phase changes at the glass/copper interface. The outcome is local melting followed by solidification, resulting in the formation of the joint. Optimizing the focal plane position is critical. Placing it directly on the glass or the glass/copper interface causes significant laser energy loss due to transmission. However, placing the focal plane too far from the interface results in an expanded region where a sufficiently sized and effective melt zone cannot be formed at the interface. Following multiple experimental tests, a defocus distance of 0.5 mm from the interface was identified as appropriate for our study's laser system.

In Fig. 2, an optical microscope (OM) image presents a well-formed weld seam between the glass substrate and the copper plate was formed. The welding process was executed under the parameters $P = 15$ W, $v = 40$ mm/s, and $N = 5$. Notably, upon close examination, there are no apparent defects visible to the naked eye, as depicted in the inset for section A. This highlights the capability of the current laser processing system to produce a well-joined weld seam. However, on the right side of the welding line, Newton's rings are discernible, characterized by three distinct colors (green, blue, and pink). This occurrence underscores the challenge of achieving optimal optical-contact conditions during the weld formation [13], despite the application of clamping force throughout the welding process.

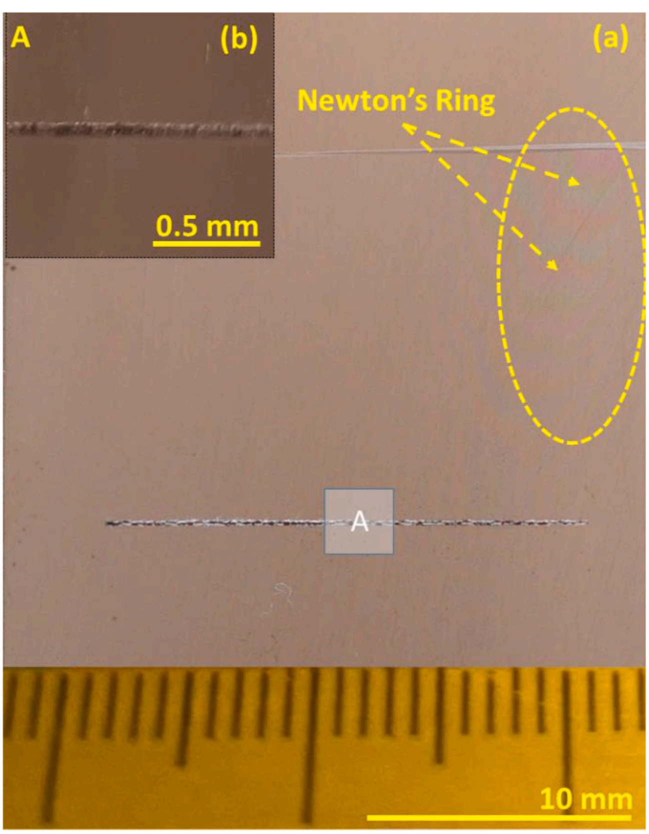

Fig. 2. Optical microscope (OM) images from the CLSM demonstrating the successful attainment of line welded seams with the following processing parameters: $P = 16$ W, $v = 40$ mm/s, $f = 20$ kHz, and $N = 5$. (a) Top view of a 20 mm-long line weld seam. (b) A magnified CLSM image showcasing the excellent surface morphology achieved.

3.2. Microstructure and elemental compositions

3.2.1. Micro- and nano-structures

In Fig. 3, the SEM images present cross-sectional views of the copper/glass weld seam were examined. The processing parameters of $v = 40$ mm/s and $N = 5$ were fixed, while $P = 15$ W ($L = 0.375$ J/mm) and 16 W ($L = 0.400$ J/mm) were set for Fig. 3 (a) and 3 (b), respectively. Fig. 3(a) showcases the absence of significant defects, such as pores or cracks, within the weld zone, aligning with the overall weld structure depicted in Fig. 2. This cross-sectional image highlights that the predominant composition of the weld zone was glass. Two primary factors contribute to this observation. First, the glass transition temperature is lower than the melting point of copper. As laser-irradiated energy was absorbed by the copper surrounding the glass/copper interface, the local copper temperature increased rapidly, leading to energy transfer to both the glass and another portion of the copper via heat conduction. This process resulted in the softening of the glass and subsequent melting of both glass and copper around the interface. Second, the incident laser originating from the glass side, coupled with the nonlinear optical absorptivity of high-temperature softened glass, played a role in extending the molten zone of the glass. As the glass temperature increased, its opacity to incident laser light intensified, causing enhanced absorption by the glass and further expansion of the molten zone in the glass.

As depicted in Fig. 3 (a), the boundary between glass and copper in the weld zone is clearly identifiable, with the glass side in the upper portion and the copper side in the lower part. The distinct demarcation of the joint boundary indicates an improvement in the Cu/SiO2 joint, highlighting a lack of micro-defects at the FIB location. Notably, several visible copper spots were observed on the glass side. In contrast, minimal glass was evident on the copper side, except in the vicinity around the interface, as highlighted by the red dashed enclosure and its magnified inset in Fig. 3(a_1). This could be attributed to several reasons: Firstly, during laser irradiation, the absorbed energy facilitated rapid local temperature elevation on the copper surface. This conduction process was repeated until the melts of glass and copper.

During this phase, the dominant mechanism for mixing within the molten zone was convectional intra-mixing, driven by differences in composition gradients and local pressure differences [17]. Other frequently discussed mechanisms for enhancing mixing in the weld pool include the Marangoni effect [13,18] and thermal buoyancy. However, in the context of this study, these mechanisms were relatively less influential. The primary reason is that the Marangoni effect is primarily triggered by temperature gradients at the interface between different phases. In scenarios characterized by low mutual solubility, such as at the interface between an exposed melt pool and air, this effect might conceivably come into play. However, the molten copper, capable of diffusing and blending with the molten glass, diminished the significance of the interface effect. Furthermore, owing to the small size of the molten pool (at the microscale) and the high viscosity of both the liquid metal and molten glass, generating significant convection currents for mixing in this study posed a challenge for thermal buoyancy.

In a molten zone comprising melted copper and glass, copper exhibited greater ease of movement or diffusion compared to silicon. This behaviour can be attributed to several factors. Firstly, there is a difference in atomic sizes, with Cu atoms being larger than silicon atoms. The larger atomic size of copper enables it to move more freely within the molten material, occupying interstitial positions and diffusing through the liquid more easily. On the other hand, the smaller size of silicon atoms restricts their mobility, making it more challenging for them to diffuse within the molten zone. Secondly, there is a discrepancy in diffusion coefficients, with Cu typically having a higher diffusion coefficient compared to Si in liquid or molten states. The diffusion and migration structure of glass and metal during laser transmission welding constituted a dynamic process [12]. The temperature gradient in the focal area resulted in the diffusion of heat, leading to the migration of elements. Thermodynamic factors contributed to different diffusion

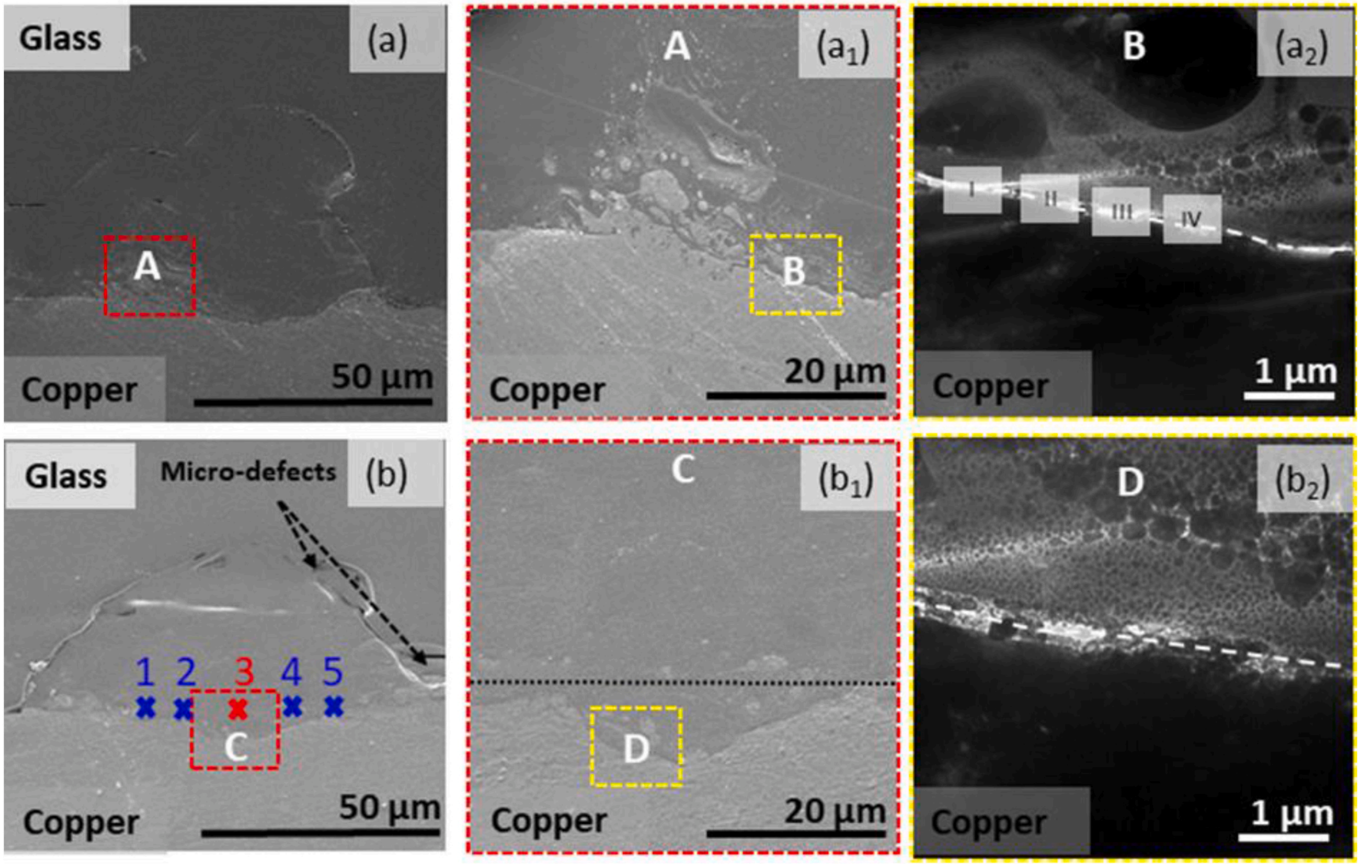

Fig. 3. Cross-sectional SEM surface morphologies illustrating internal microstructures within the weld seam, produced using the processing parameter sets of $P = 15$ W, $v = 40$ mm/s, $f = 20$ kHz, and $N = 5$ for (a), and $P = 16$ W, $v = 40$ mm/s, $f = 20$ kHz, and $N = 5$ for (b). Fig. 3(a_1) provides a higher-resolution SEM image for the enclosed area A in Fig. 3(a), revealing the mixing of copper and glass. In Fig. 3(a_2), a more detailed observation of the periphery of the interfacial glass/copper boundary in the B zone indicated in Fig. 3(a_1) is presented. Similarly, Fig. 3(b_1) and Fig. 3(b_2) present enlarged SEM images for the enclosed C area in Fig. 3(b) and the D region in Fig. 3(b_1), respectively. The locations from (1) to (5) in Fig. 3(b) denote specific positions for conducting XPS measurements. Additionally, red markings confirm the precise location (3) for determining the XPS spectrum, as depicted in Fig. 5. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

behaviours, with copper exhibiting a stronger affinity for oxygen compared to silicon. In the presence of molten SiO_2, copper readily reacts to form copper oxide compounds, promoting the diffusion of copper within the molten zone by creating additional pathways for copper atoms to move through the material [14].

In Fig. 3(a_2), a distinct interface, separate from the native copper region and delineated by the dashed line, is evident. In areas surrounding this demarcation, copper demonstrated its capability to establish chemical bonds with oxygen from the glass at elevated temperatures. Consequently, Cu^{+} or Cu^{2+} compounds were formed, as confirmed by HR-XPS analysis as discussed in Section 3.2.2. Depending on the instantaneous local temperature fluctuations along this boundary, a variety of binding compositions might arise, leading to the emergence of various Cu^{+}-O or Cu^{2+}-O species, ultimately resulting in the formation of Cu^{+}/Cu^{2+}-O-Si compositions (referred to as Cu^{x+}-O-Si compound). This plays a crucial role in enhancing the bonding strength between Cu and SiO_2, as outlined in Ref. [19]. A more detailed investigation of the nanostructure within this region will be undertaken subsequently through HR-TEM observations, as illustrated in Fig. 4.

Upon increasing the incident laser line intensity to 0.4 J/mm in Fig. 3 (b), noticeable changes occurred in the glass/copper interfacial region, which expanded and acquired a smoother and more uniform appearance concerning both the interface line and the homogeneity of its periphery mixing. In Fig. 3(b_1), the previously discernible aggregated copper spots seen in Fig. 3(a_1) were notably resolved due to stronger mixing and a prolonged duration of the molten state caused by the higher incident laser energy. Comparing Fig. 3(b_2) to Fig. 3(a_2), a larger and brighter zone surrounding the dash-line indicated interface is observed, indicating that higher laser energy facilitates the penetration of copper micro-particles into the glass, thereby promoting complete reactions among copper, oxygen, and silicon [13,14]. Consequently, one of the key determinants of weld quality hinges on whether the laser provides locally sufficient and appropriate energy to enable a thorough redox reaction between Cu and SiO_2. In the subsequent stages of this study, we will explore two approaches to adjusting laser energy: modifying the laser line energy intensity and regulating the total number of scans. Through scrutinizing the effects of these variations on bond strength, our objective is to enhance our understanding of this crucial issue.

As depicted in Fig. 3(a_2), along the dash line, four distinct regions labelled I, II, III, and IV were subjected to FIB slicing, and the outcomes are illustrated in Fig. 4(a) and (b), 4(c) and 4(d), respectively. In Fig. 4 (a), the differentiation between glass- and copper-rich sides was discernible based on the grayscale intensity, distinguishing lighter and darker regions. A single crystalline structure, Cu (111), corresponding to a d-spacing of 0.208 nm and aligning with the face-centred cubic structure of Cu (JCPDS-IDDD no.71-4610, $d_{111} = 0.2089$ nm), is evident. Moving to Fig. 4(b), structure of copper oxide, Cu_2O (111), featuring a d-spacing of 0.245 nm is clearly identified. This value aligns well with the Cu_2O face-centred cubic structure (JCPDS-ICDD no. 65-3288, $d_{111} = 0.2463$ nm). Meanwhile, Fig. 4 (c) reveals the presence of cupric oxide, CuO (111) and cuprous oxide, Cu_2O, exhibiting a d-spacing of 0.232 nm, 0.246 nm, respectively. These oxides manifested due to the proximity of

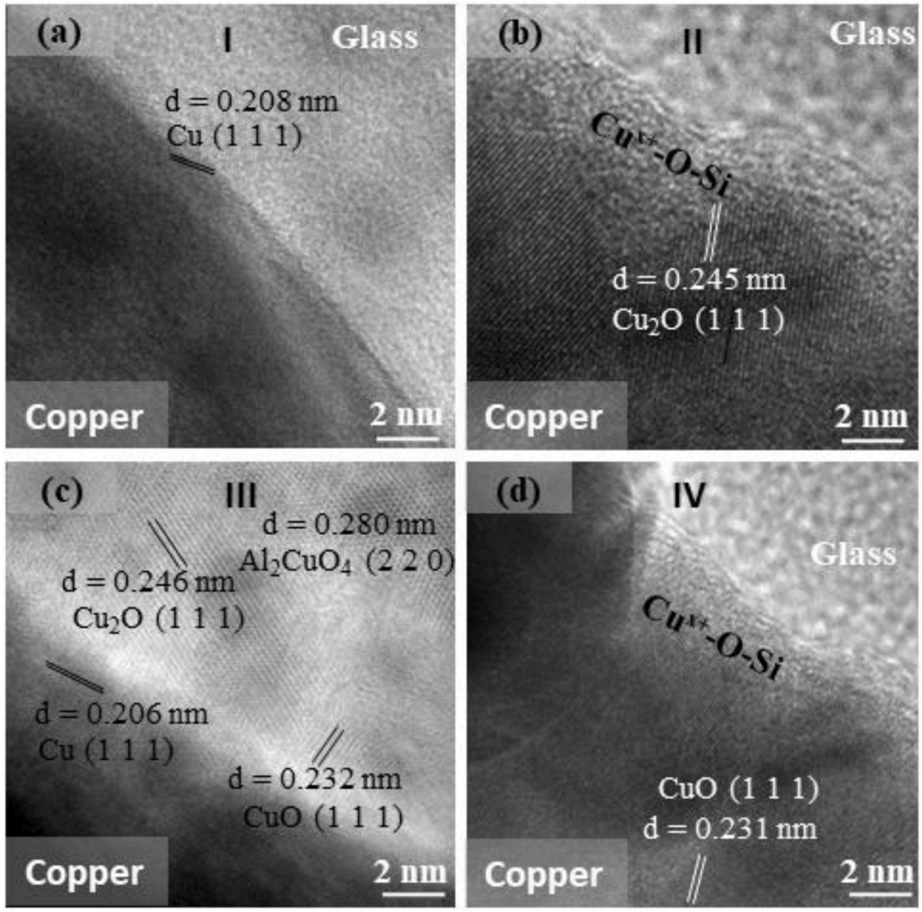

Fig. 4. High-resolution TEM images (a), (b), (c), and (d) depict the four indicated glass/copper interfacial regions I, II, III, and IV highlighted in Fig. 3(a_2), respectively.

copper to the glass, forming bonds with the oxygen within the silica matrix. Additionally, the emergence of the Al_2CuO_4 phase was observed, potentially attributed to the initially high Al_2O_3 content (17.11%) in the EAGLE® glass, becoming prominent at elevated temperature during the welding.

In Fig. 4(d), CuO (111) is evident, with a corresponding d-spacing of 0.231 nm (JCPDS-ICDD no. 45-0937, $d_{111} = 0.23$ nm). Furthermore, the region of Cu^{x+}-O-Si is prominent in Region IV, and a higher distribution of copper nanoparticles in this region, compared to Region III, resulted in a more obvious and thicker layer of Cu^{x+}-O-Si. Consequently, dynamic reactions driven by the incident laser energy occurred at the Cu/SiO_2 interface during the welding process.

The observable transformation in the appearance of the new compound Cu^{x+}-O-Si, occurring at the juncture of copper nanoparticles and amorphous glass across the majority of the weld's width, provides compelling evidence of a potent chemical reaction within the weld's constituents. Nonetheless, the dimensions of the freshly formed compounds exhibit variability across different regions of the Cu/SiO_2 interface [17].

3.2.2. Elemental composition

A crucial requirement for achieving better welding performance is the formation of strong chemical bonds among elements within the weld zone. In this investigation, an XPS test was employed to examine the elemental composition, chemical state, and electronic state caused by chemical interactions among the elements Cu, O, and Si. Following the methodology outlined in Ref. [17], measurements were averaged from five distinct central positions on the cross-sectional plane of the weld zone, perpendicular to the line weld seam, as depicted in Fig. 3(b). X-ray measurements were conducted with a spot size of 400 μm. In this investigation, the XPS spectrum was acquired near position No. 3, as indicated in Fig. 3(b), aligning with the results presented in Fig. 5(a), (b), (c), and (d). Fig. 5(a) illustrates the survey scan of the samples obtained using the Al Kα beam. At the Cu/SiO_2 interface, peaks corresponding to the spectrum lines of Cu 2p, O 1s, Si 2p, and C 1s (attributed to carbon contamination) were observed, with binding energies approximately at 934, 530, 101, 76, and 284 eV, respectively [20].

In Fig. 5(b), the observed peak at 934.9 eV was attributed to Cu $2p_{3/2}$, with possible contributing sources including metallic Cu, Cu^+, Cu^{2+} oxide phases [21], and Cu–Si combination [22]. The heavy overlap of several peaks made it challenging to distinguish them from the XPS data. However, the robust and broad signals of peaks suggested the coexistence of these two species, including Cu^0 and Cu^+ [23]. Notably, the band at 931.5 eV is attributed to the Cu^{x+} state in Cu $2p_{3/2}$, which has energies 3.4 eV lower than Cu^0/Cu^+/Cu^{2+}/Cu–Si phases. At the bonding energy, shake-up satellite peaks were observed at approximately 10 eV and 7.6 eV higher than the $2p_{1/2}$ and $2p_{3/2}$ peaks, respectively [24,25]. Meanwhile, the peak extraction at 952.5 eV for Cu^0/Cu^+ species corresponds to Cu $2p_{1/2}$ [26,27].

Continuing the analysis of the distinctive elemental characteristics of O 1s, with a binding energy of 531.46 eV, as depicted in Fig. 5 (c), and considering the notable abundance of oxygen within the glass family, the prominent peak at 532.5 eV was attributed to the O–Si combination [28,29]. Simultaneously, the peak extraction at 530.6 eV is associated with the formation of copper oxides, predominantly Cu_2O [30]. The O^x state, specifically represented by the peak centred at 531.7 eV, may result from the sample's Cu–O–Si mixed oxide state forming [31]. At the bonding energy, shake-up satellite peaks were observed at

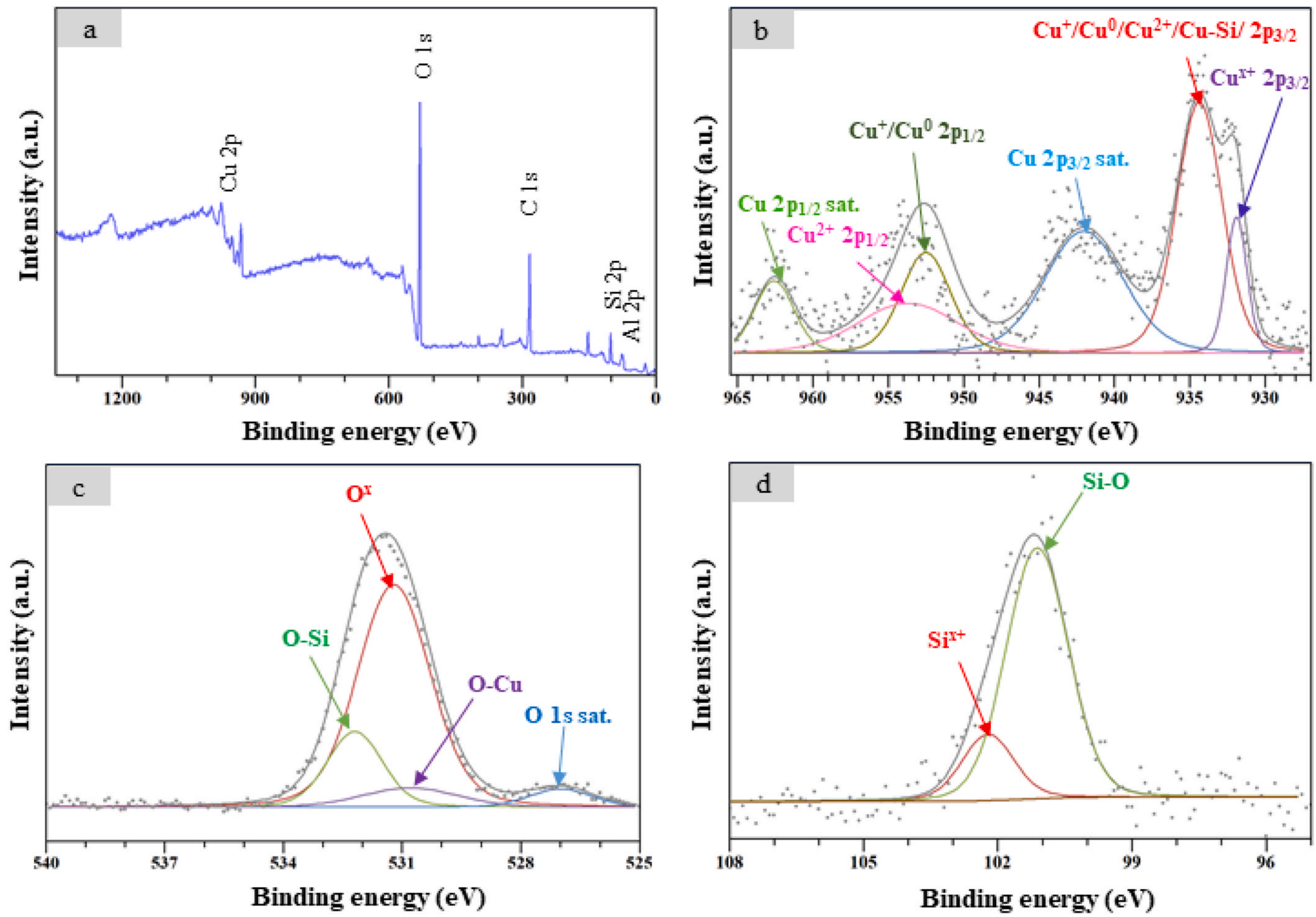

Fig. 5. High-resolution XPS spectra, including the full survey spectrum (a), Cu 2p spectrum (b), O 1s spectrum (c), and Si 2p spectrum (d). These spectra correspond to the processing parameter set: $P = 16$ W, $v = 40$ mm/s, $f = 20$ kHz, $L = 0.400$ J/mm, and $N = 5$. The grey dots represent measured data points, while the grey lines depict the fitting lines generated using CasaXPS software. The corresponding compositions identified from Cu 2p, O 1s, and Si 2p are indicated by arrows.

approximately 3.4 eV lower than the O–Cu peaks. In Fig. 5 (d), the high-resolution line of Si 2p exhibits a double extraction peak at 101.1 eV and 102.7 eV, corresponding to Si–O and Si_x, respectively [32].

The XPS spectrum results reveal that Cu 2p, O 1s, and Si 2p represented significant states, encompassing the Cu^{x+}-O state and Cu^{x+}-O-Si state, which manifest on the Cu/SiO_2 interface [19]. Additionally, the interface region exhibited the presence of Cu^{x+}, $O^{x,}$ and Si^{x+}, foretelling the formation of new phases that played a pivotal role in enhancing bonding preformation [13,31]. This is crucial in understanding the chemical bonding mechanism between Cu and SiO_2 and underscores the importance of the emerging phases in the bonding process.

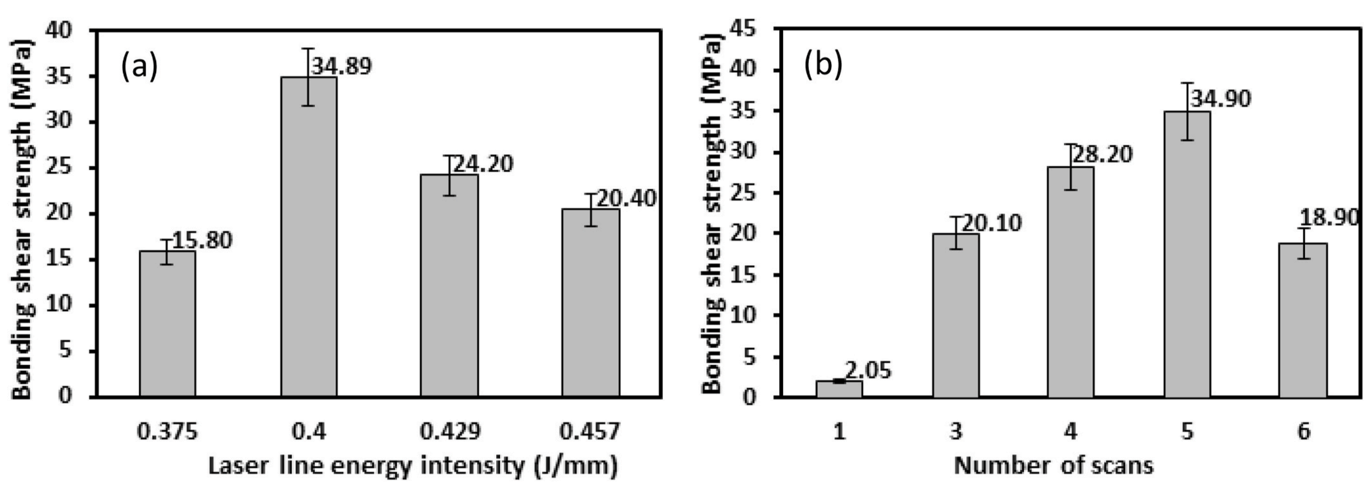

Fig. 6. Shear force separation testing was employed to measure bonding strength as a function of laser line energy intensity (a) and the number of scans (b). The laser line energy intensity varied from $L = 0.375$ J/mm to 0.457 J/mm, with the number of scans adjusted at $N = 1$, 3, 4, 5, and 6.

3.3. Bonding strength

The shear force separation test was utilized to evaluate the strength of the welded copper/glass bond. The bond strength was determined using formula $\sigma = \frac{F_{max}}{S}$, where $F_{\max}$ represents the maximum shear force leading to the separation failure of the weld, and S is the projected surface area of the welded seam [17]

The first situations examined were that illustrated in Fig. 6 (a). The processing conditions were through maintaining a frequency of $f = 20$ kHz and a fixed number of repeated scans $N = 5$. The power levels chosen were $P = 15$ W and 16 W, accompanied by scanning speeds of $v = 40$ mm/s and 35 mm/s. The resulting shear strengths for these cases are depicted in Fig. 6 (a), illustrating values of $\sigma = 15.80$, 34.89, 24.20, and 20.40 MPa for laser line energy intensities of $L = 0.375$ J/mm, 0.400 J/mm, 0.429 J/mm, and 0.457 J/mm, respectively.

The results unveiled an initial weld strength of $\sigma = 15.80$ MPa, which significantly increased by more than twofold to 34.89 MPa with the augmentation of L from 0.375 J/mm to 0.400 J/mm. However, as L was further increased to 0.429 J/mm, σ exhibited a noticeable reduction to 24.2 MPa. This reduction persisted with the increase in L to 0.457 J/mm, resulting in a shear strength of 20.4 MPa. Notably, the highest bonding strength achieved in this study surpassed that of 33.4 MPa, as demonstrated in Ref. [13], where glass and copper direct bonding was also accomplished using a ns laser system. The variations in the bond strength are attributed to factors associated with microstructures within the weld zone and its periphery, as illustrated in Fig. 7.

Figs. 7 and 8 depict the cross-sectional surface morphologies of the weld seam, captured using CLSM and obtained under different processing parameters. The processing parameters for Fig. 7(a)–7(d) corresponded to the four cases illustrated in Fig. 6(a), respectively. Similarly, those for Fig. 8 (a), 8 (b), 8 (c), 7 (b), and 8 (e) corresponded to the five situations depicted in Fig. 6(b).

In Fig. 7(a), it is evident that the interfacial span of the glass/copper joint was notably limited. Additionally, conspicuous defects along the interface played a significant role in yielding a low joint strength. In contrast, as shown in Fig. 7(b), the joining span was noticeably enlarged, and defects in this region were significantly reduced. This observation clarified the substantial disparities in the bond strength between the cases obtained from $L = 0.375$ J/mm and 0.400 J/mm, respectively. Moving on to the case of L at 0.429 J/mm, referring to Fig. 7(c), although the joining span remained large, the defects along the interface were enlarged as well, resulting in a decline in the bond strength. Further increasing L to 0.457 J/mm, the additional reduction in the bond strength could be attributed to the formation of more defects scattered in the interfacial region and the very irregular boundary along the glass/weld zone interface, as depicted in Fig. 7(d).

The bond strength of the weld under the processing conditions $P = 16$ W, $v = 40$ mm/s, and $f = 20$ kHz is illustrated in Fig. 6(b) for various total scan numbers, $N = 1$, 3, 4, 5, and 6. It is noteworthy that the initial bond strength was relatively low, approximately 2.05 MPa at $N = 1$, but underwent a significant enhancement to $\sigma = 20.10$ MPa after two additional post-processing treatments. As N increased to 4 and 5, the bond strengths experienced further improvement, reaching 28.2 and 34.9 MPa, respectively. However, at $N = 6$, σ declined to 18.9 MPa. Analysis of Fig. 8(a) revealed evident defects along the glass/copper interfacial zones, and the boundary at the glass/weld zone exhibited a waviness, lacking of smoothness. Consequently, this configuration failed to demonstrate a convincing microstructure that would contribute to a robust bond strength. Post-processing interventions played a crucial role in ameliorating the defects along the interfacial area, and the periphery of the weld zone, especially the boundary of the glass/weld zone was

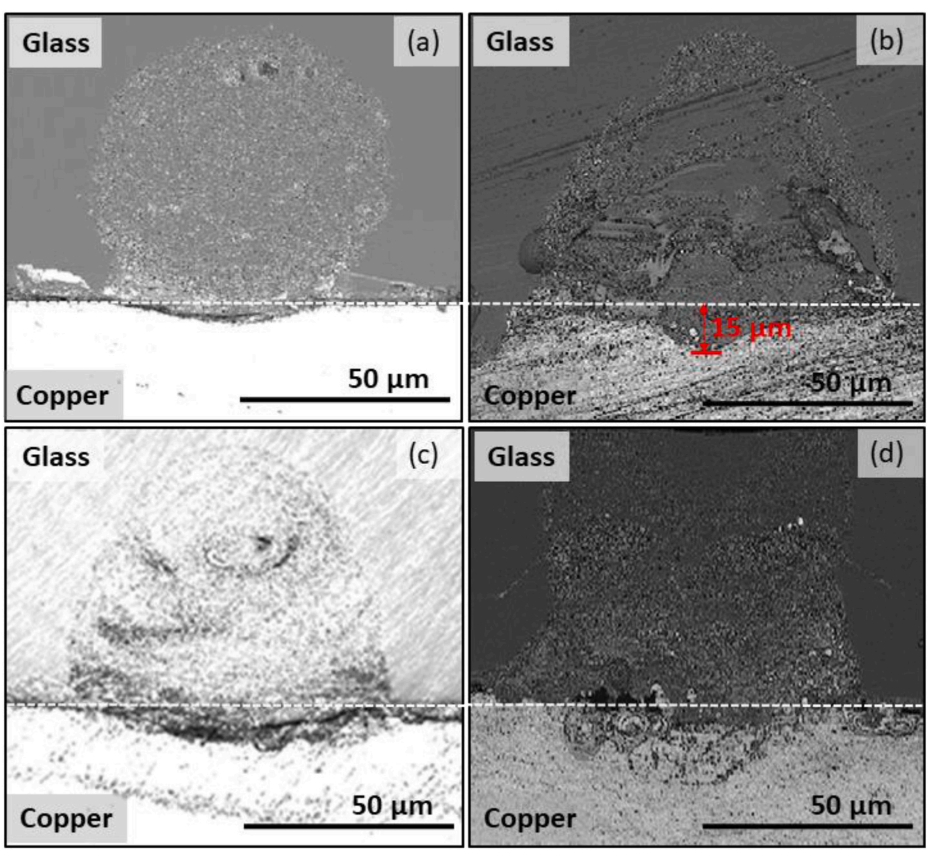

Fig. 7. Confocal laser scanning microscope images depicting cross-sectional surfaces of the weld seam at varying laser line energy intensities. The processing parameter sets were as follows: (a) $L = 0.375$ J/mm ($P = 15$ W, $v = 40$ mm/s); (b) $L = 0.400$ J/mm ($P = 16$ W, $v = 40$ mm/s); (c) $L = 0.429$ J/mm ($P = 15$ W, $v = 35$ mm/s); and (d) $L = 0.457$ J/mm ($P = 16$ W, $v = 35$ mm/s). In all cases, the pulse repetition rate and number of scans were fixed at $f = 20$ kHz and $N = 5$, respectively.

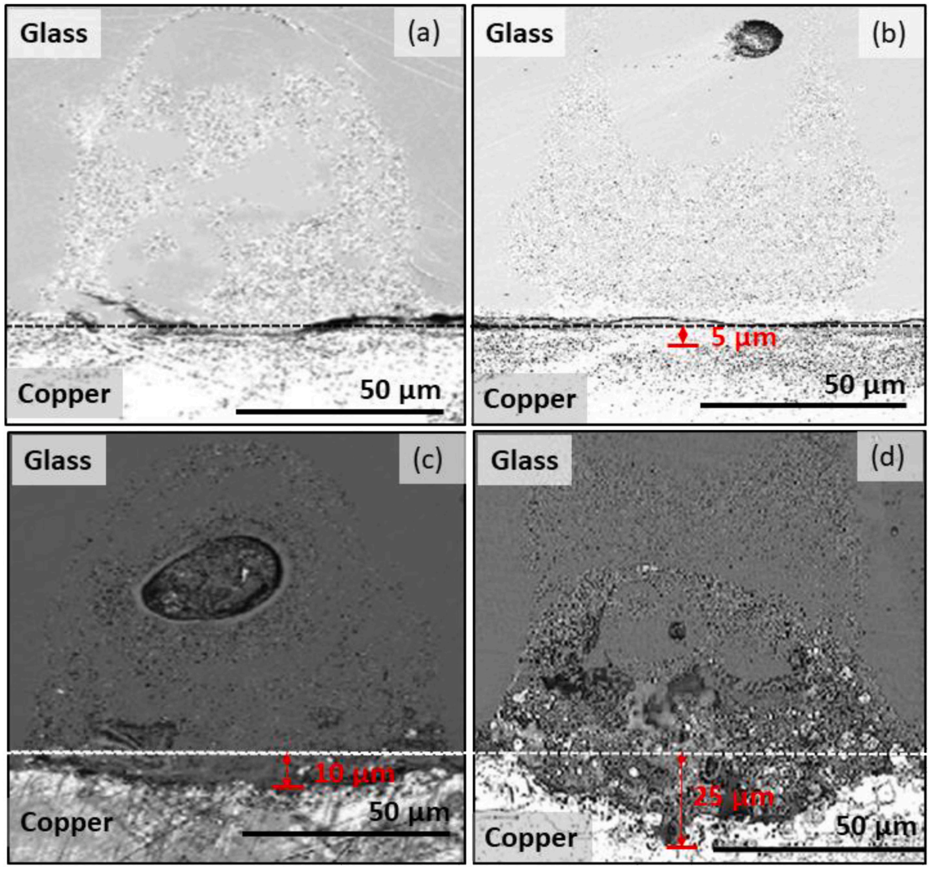

Fig. 8. Confocal laser scanning microscope images illustrating cross-sectional surfaces of the weld seam with different numbers of scans: $N = 1$ for (a), $N = 3$ for (b), $N = 4$ for (c), and $N = 6$ for (d). The processing parameters were held constant at $L = 0.4$ J/mm ($P = 16$ W, $v = 40$ mm/s), and $f = 20$ kHz.

visibly smoother, as depicted in Fig. 8(b) and (c) and. At $N = 5$, as shown in Fig. 7(b), the defects at the interfacial region and non-smooth boundary problems were significantly improved, resulting in an obvious enhancement in bond strength. However, at $N = 6$, Fig. 8(d) indicates that the microstructures, in the aforementioned two main aspects, deteriorated again, leading to a decline in bond strength.

3.4. *Influence of processing parameters on the weld characteristics*

The cross-sectional surface morphology of the weld seam, as shown in Figs. 7 and 8, is re-examined to assess the influence of two focused processing parameters, namely laser line energy intensity and total scan number, on the shape of the weld zone and its corresponding characteristics.

3.4.1. *Effect of laser energy*

Since the majority of the weld zone consisted of the modified glass region, softened during the welding and re-solidified after the welding process. The cross-sectional images reveal variations in the shapes of the re-solidified glass region, contingent upon changes in incident line energy intensity. In Fig. 7(a)–a shell shape is evident, transitioning to a hump shape in both Fig. 7(b) and (c) and , and finally to an M shape in Fig. 7(d). As previously noted, the incident laser energy primarily absorbed by the copper at the glass/copper interface. At a lower incident laser energy level, the higher energy absorption along the incoming laser incident path was not as pronounced. Nevertheless, the distribution of heat from the interfacial area through diffusion remained significant, resulting in the diffusive spread of heat from the interface into the surrounding areas and shaping a more circular, shell-like re-solidified glass region.

With an increase in incident laser line energy intensity, not only did the glass/copper interfacial region expand, but the height of the re-solidified glass region also increased. The lengthening of the interfacial line resulted from a broader range of direct laser incident energy absorption around the interfacial region. Simultaneously, the height increase attributed to the higher temperature-induced enhancement in absorption coefficient and opacity along the laser beam's incoming path. The widening of the interfacial line and elongation in height contributed to the formation of a hump shape in the re-solidified glass region. Moreover, as the irradiated laser line energy intensity further increased, the absorption along the incoming beam passage become more significant, causing hot spots to elevate. This elevation resulted in their surroundings experiencing higher temperatures and pressures, generating extra expansion momentum along the beam passage and aiding in the formation of the M-shaped re-solidified glass area.

3.4.2. *Influence from number of scan*

As previously highlighted, the cumulative number of scans emerged as a pivotal focal point in this investigation, akin to post-processing. Its impact, contingent upon the appropriateness of the applied treatment, could either augment or diminish bonding performance. In this context, we explored the intricate correlations among cross-sectional morphologies of weld seams, bonding strength, and the overall number of scans. Fig. 8(a) and (b), 8(c), 7(b), and 8(d) exhibit the cross-sectional surface morphologies corresponding to scan numbers $N = 1$, 3, 4, 5, and 6, respectively, while maintaining fixed processing parameters of P at 16 W and v at 40 mm/s.

In Fig. 8(a), the initial surface morphology arising from $N = 1$ is depicted, showcasing noticeable defects along the glass/copper interface. Additionally, defects were evident in the re-solidified glass region near the interface. Moreover, distinct inhomogeneity was observed within the re-solidified glass region, accompanied by an irregular periphery along the boundary of the region. These observations were attributed to limitations in the driving momentum of mixing and diffusion, hindering the achievement of uniformity. These limitations, in turn, arised from constraints in both the amount and duration of energy supplied.

Following two additional scans at $N = 3$, as illustrated in Fig. 8(b)–a significant reduction in defects around the glass/copper interface areas was evident. Moreover, there was an enhancement in the surface morphology of the re-solidified glass region. However, an inhomogeneous area, resembling a small cavity, manifested in the upper portion of the region. Owing to its glassy structure and the presence of copper, the transparency of the re-solidified glass region to subsequent processing scans was diminished, resulting in increased absorption of laser energy in the upper portion. This surplus energy facilitated structure and ingredient redistribution, occasionally leading to uneven structural aggregation due to initial local inhomogeneity.

With the scan numbers up to 4, as depicted in Fig. 8 (c), the additional energy further promoted the formation of the cavity-like inhomogeneous area. It is important to note that, due to experimental constraints, Fig. 8(b) and (c) and were transversally cut based on samples from different batches of experiments. Consequently, the exact initial locations of the visible cavity-like inhomogeneous region could vary slightly. Nevertheless, the formation of aggregated inhomogeneity in the upper portion was consistently repeatable.

In Fig. 7(b), it is evident that both the defects around the glass/copper interfacial region and the pronounced uneven inhomogeneous structures in the re-solidified glass region were significantly reduced through repeated scans at $N = 5$, resulting in an enhanced bonding strength. However, upon increasing the scan number to 6, Fig. 8(d) reveals not only the emergence of numerous micro-holes around the interfacial area but also the presence of abundant inhomogeneous structures and microscale cavities within the weld zone of the re-

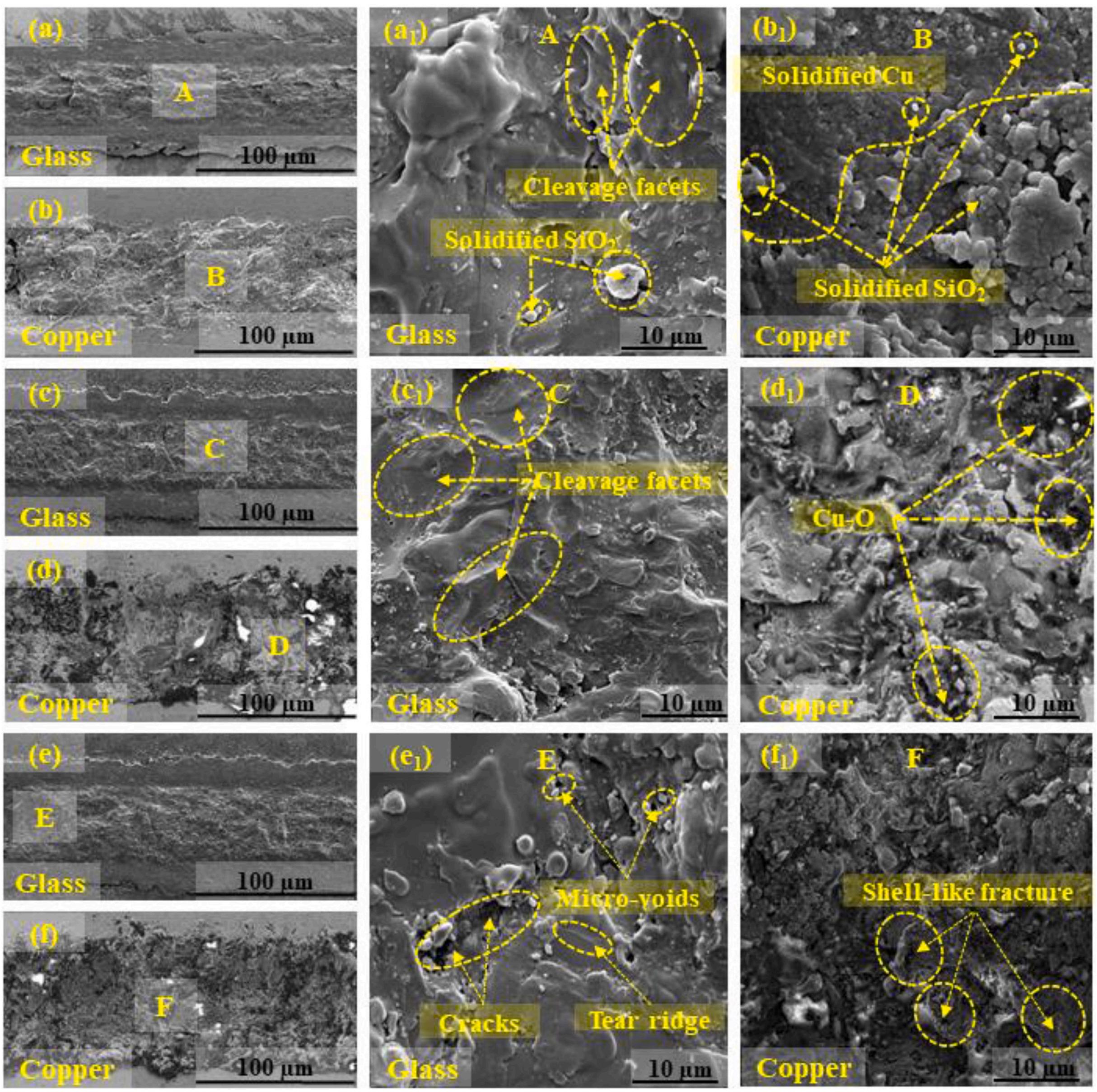

Fig. 9. SEM images revealing cleaved surfaces following shear force separation. Figures (a) and (b), (c) and (d), and (e) and (f) correspond to welds produced with various laser line energy intensities of $L = 0.375$ J/mm ($P = 15$ W, $v = 40$ mm/s), 0.400 J/mm ($P = 16$ W, $v = 40$ mm/s), and 0.457 J/mm ($P = 16$ W, $v = 35$ mm/s), respectively. Other processing parameters were kept constant at $f = 20$ kHz, and $N = 5$. Figures (a), (c), and (e) depict the surfaces on the separated glass side, while Figures (b), (d), and (f) show the surfaces on the copper side. Additionally, Figures (a_1), (b_1), (c_1), (d_1), (e_1), and (f_1) represent High-Resolution SEM (HR-SEM) images of regions A, B, C, D, E, and F, respectively.

solidified glass region. Additionally, the appearance of the M-shape re-solidified glass region, as depicted in Fig. 7(d)–as became observable at the scan numbers of $N = 6$. Consequently, post-processing the weld zone through an appropriate number of repeated scans could introduce excess energy, potentially damaging the microstructure in the weld zone and adversely affecting the resulting weld quality.

Expanding the joint glass/copper surface area, as observed from the cross-sectional surface perspective, entails increasing the span of the glass/copper joint interface and improving the microstructural quality of the weld zone-both pivotal indicators of enhanced joint strength. As discussed in the previous paragraph, post-processing through repeated scanning has been demonstrated to modify the microstructure in the weld zone, and this method has the potential to simultaneously augment the joint surface area. In Fig. 8(b) and (c), 7(b), and 8(d), the farthest distance from the original glass/copper interface, denoted as *d,* serves as an indicator of the change in the length of the joint surface resulting from variations in the *N* value. The results revealed that for $N = 3$, 4, 5, and 6, the corresponding *d* values were 5, 10, 15, and 25 μm, respectively. In this study, when $N = 5$ underwent post-processing through repeated scans, not only was the microstructure within the weld zone improved, but the glass/copper joint area was also expanded.

3.5. Fracture surface morphology

SEM images in Fig. 9(a) to 9(f) present distinct surface morphologies on both the glass and copper sides following the application of shearing separation force to the welds. With fixed processing parameters of $f =$ 20 kHz and $N = 5$, the varying parameters included power and scan speed. Specifically, for Fig. 9(a) and (b), $P = 15$ W and $v = 40$ mm/s, resulting in $L = 0.375$ J/mm; for Fig. 9(c) and (d), $P = 16$ W and $v = 40$ mm/s, yielding $L = 0.4$ J/mm; and, for Fig. 9(e) and (f), $P = 16$ W and $v = 35$ mm/s, leading to $L = 0.457$ J/mm. Fig. 9(a), (c), and 9(e) focus on separated surface morphologies on the glass substrate, while Fig. 9(b) and (d), and 9(f) highlight separated surface morphologies on the copper substrate. Each figure includes a magnified view to facilitate detailed observations, resulting six enclosed areas, designated as A, B, C, D, E, and F for Fig. 9 (a), (b), (c), (d), (e), and(f), respectively.

In the presence of a shear separation force, the fracture manifests along the weld/copper boundary, resulting in a convex surface on the glass side (see Fig. 9(a) and (c), and 9(e)) and a concave surface on the copper side (refer to Fig. 9(b) and (d), and 9(f)) [17]. High-magnification micrographs illustrating areas A and B in Fig. 9(a) and (b) are presented in Fig. 9(a_1) and Fig. 9(b_1), respectively. As depicted in Fig. 9(a_1), the majority were the discernible cleavage facets, while secondarily includes solidified SiO_2 indicating a clear manifestation of brittle fracture. Conversely, the fracture surface morphology in region B of Fig. 9 (b_1), concave copper side, predominantly consists of solidified SiO_2, represented by a lighter-color area. Also, under impact of lower laser line energy intensity of $L = 0.375$ J/mm, the Cu^{x+}-O-Si compound could not be obviously formed. Consequently, the area that remains could be the location of solidified Cu, as indicated by the grey area at the top of the field. Conversely, the fracture surface morphology in region B of Fig. 9(b_1), representing the concave copper side, predominantly consists of solidified SiO2, represented by a lighter-colored area. Also, under the impact of lower laser line energy intensity of $L = 0.375$ J/mm, the Cu^{x+}-O-Si compound could not be obviously formed. Consequently, the remaining area could be the location of solidified Cu, as indicated by the grey area at the top of the field. These findings suggest that the separation was primarily achieved through brittle fracture.

With an increase in laser line energy intensity to $L = 0.400$ J/mm, in comparison to Fig. 9(a_1), the fracture surface morphology depicted in Fig. 9(c_1) reveals the occurrence of additional cleavage planes. The heightened laser processing energy fostered a more robust mixing and diffusion process around the interfacial region, resulting in less distinct boundaries between cleavage facets. Simultaneously, the fracture morphology on the copper side (Fig. 9(d_1)) exhibits the presence of Cu–O compounds, denoted by darker regions amidst the predominantly brighter area (Cu^{x+}-O-Si compounds). In contrast to Fig. 9(b_1), the elevated laser line energy intensity facilitated the formation of connections between Cu and SiO_2. The emergence of copper oxide, cuprous oxide, (and Cu^{x+}-O-Si compounds) contributed to a less identifiable re-solidified SiO_2 area.

As the laser line energy intensity was elevated to $L = 0.457$ J/mm, Fig. 9(e_1) depicted a typical fracture surface morphology on the glass side characterized by brittle fracture, evident from the tear ridge characteristics. Additionally, various micro-defects, such as cracks and micro-voids, became apparent, exerting a detrimental influence on bonding quality and resulting a decline in bond strength shown in Fig. 6 (b). Concurrently, the heightened laser incident energy resulted in the expansion of oxidized copper zones, discernible as darker regions in Fig. 9(f_1). Notably, the presence of a shell-like fracture on copper, enclosed by the dash line at the bottom portion, indicated the identification of glass debris at elevated laser line energy intensity. In comparison to Fig. 9(b_1) and 9(d_1), the formation of the shell-like fracture emerges as a critical factor influencing brittle fracture and adversely affecting bonding performance.

4. Conclusion

In this study, a nanosecond (ns) laser was utilized for the laser transmission welding of EAGLE XG® glass and C1100 copper. The investigation focused on analyzing the macrostructure, microstructure, chemical composition, phase composition, nanostructure of the interface region, and mechanical properties, including the observation of fracture surfaces in copper/glass joints. The conclusions drawn from this study, in correspondence to the results and discussion section, are as follows.

1. The investigation streamlined and defined the suitable range of welding parameters, or the processing window for our system, based on the ns laser bonding mechanism. The typical macrostructure of the copper/glass joint exhibited a well-formed weld seam between the glass substrate and the copper plate without apparent defects visible to the naked eye. Additionally, the microstructure of the copper/glass joint revealed an ideal region with a uniform bonding line.
2. The mixing mechanism of copper and glass was explored in relation to the effect of compositional gradients and local pressure differences. This resulted in the non-uniform presence of observable copper and glass spots in the molten zone. XPS and HR-TEM observations during inter-diffusion and intra-mixing processes revealed the presence of Cu^{x+}, Si^{x+}, O^{x}, and Al at the joint interface, leading to chemical reactions that could generate an amorphous layer of Cu^{x+}-O-Si, critical for improving the quality of Cu/SiO2 joints.
3. The correlation between breaking shear strength, processing parameters, and microstructure of Cu/SiO_2 joints was investigated. With specific laser parameters—laser line energy intensity of 0.4 J/mm, laser power of 16 W, scanning speed of 40 mm/s, repetition rate of 20 kHz, and scanning number of 5 scans—the weld's shear strength reached approximately 35 MPa without any observed micro-defects.
4. The local temperature of copper and glass rapidly rose to the melting point with an increase in laser line energy intensity to an appropriate level. Encouraging redox reactions enhanced the development of the fusion zone, resulting in stronger weld strength with the formation of a hump shape in the re-solidified glass region. As the total number of scans increased, the absorbed heat input on the copper surface significantly rose, leading to an increase in the molten zone's depth due to the expansion of intermixing and intra-diffusion, contributing to an overall increase in shear strength.

5. Fracture surface analysis revealed that at various energy levels, the fractured glass surface was dominated by cleavage surface features, whereas the fracture morphology at the copper surface was notably influenced by the presence of various compositions, such as solidified glass, Cu–O, and Cu^{x+}-O-Si. The observation of brittle fracture samples highlighted the influence of laser line energy intensity on the bond morphology after subjecting the samples to shear testing.

CRediT authorship contribution statement

Hoai Nguyen: Data curation, Investigatio, Visualizatio, Writing – original draft. **Chih-Kuang Lin:** Validation. **Pi-Cheng Tung:** Validation. **Jeng-Rong Ho:** Conceptualization, Data curation, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Acknowledgments

The author gratefully acknowledges the sponsorships from the Ministry of Science and Technology of Taiwan under Grant Numbers of MOST 110-2221-E-008-043-MY2, MOST 111-2622-E008-018 and MOST 111-2218-E-008-006.

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